

Model selection

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Last time: Cross-validation and model selection

Crucial: We must cross-validate the whole model selection process, not just the final chosen model!

Today

- ▶ Model selection and hypothesis testing: can I perform valid inference after choosing a model? (activity and discussion)
- ▶ Cross-validation and model selection: activity

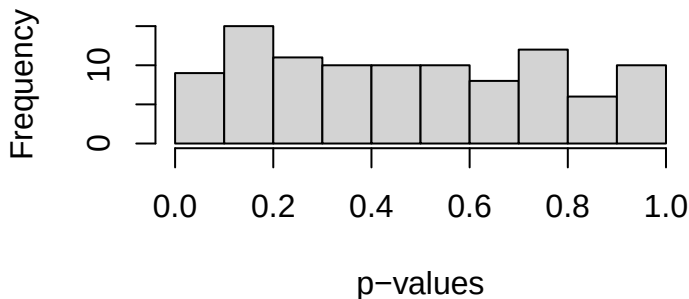
Activity 1: Model selection and hypothesis testing

Work on the first class activity (course website)

Activity 1

Distribution of reported p-values across all variables:

Histogram of p-values



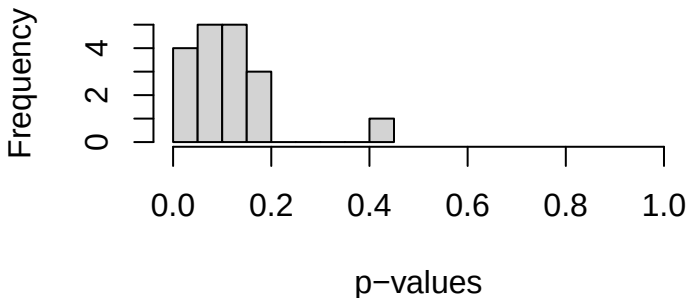
```
mean(pvals < 0.05)
```

```
## [1] 0.03960396
```

Activity 1

Distribution of reported p-values for only the *selected* variables:

Histogram of p-values



```
mean(pvals < 0.05)
```

```
## [1] 0.2222222
```

Question: Why is testing hypotheses on the selected variables problematic?

The problem

- ▶ Model selection will choose the variables that appear related to the response *in the observed data*
- ▶ If we test only the selected variables, and use the same data for selection and testing, then we have biased the outcome ahead of time: we *know* that these variables must appear related to Y in the data!
- ▶ As a result, we lose type I error control

Data splitting

Limitations of automated search methods

- ▶ Not suitable if we have particular inferential questions to test
- ▶ BSS can be computationally expensive
- ▶ Stepwise procedures are greedy and are not guaranteed to find an optimal model (or even a “good” model)
- ▶ Automated procedures **don't** do any of the following:
 - ▶ Check model diagnostics
 - ▶ Experiment with different transformations
 - ▶ Assess model performance
 - ▶ Identify unusual observations
 - ▶ etc.

Activity

Work on the activity (course website).